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Denmark: Roskilde Bank Restructuring, 2008¹

Bailey Decker²

Yale Program on Financial Stability Case Study

March 28, 2024

Abstract

Roskilde Bank A/S was the eighth-largest bank in Denmark at the time of the Global Financial Crisis, with 43 billion Danish kroner (DKK; USD 9.1 billion) in consolidated assets as of March 2008. Roskilde had considerable exposure to real estate and construction markets, prompting ratings downgrades and larger write-downs than expected. On July 10, 2008, Roskilde asked for liquidity assistance from the Danish central bank, Danmarks Nationalbank (DNB). DNB and the banking sector group, Private Contingency Association (PCA), announced emergency liquidity assistance in the form of an unlimited credit facility to Roskilde on July 10, 2008, conditioned on Roskilde's full or partial sale. When a private buyer did not emerge, DNB and the PCA took over Roskilde on August 24, 2008. Danish authorities transferred most of the assets and liabilities from Roskilde (Old Roskilde), including DKK 14.5 billion outstanding debt on the emergency loan, into a new entity: New Roskilde. DNB and the PCA then made a capital injection of DKK 4.5 billion into New Roskilde to meet regulatory solvency requirements and to protect existing depositors and other senior creditors from losses. Existing shareholders and subordinated debt holders lost an estimated DKK 5.2 billion, compensated by an "earn-out" provision that was unlikely to generate any benefit to them. On August 10, 2009, the newly created Financial Stability Company (FSC) took over New Roskilde for resolution. The resolution process took longer than Danish authorities expected when buyers of New Roskilde's assets returned a large number of loans and accompanying deposits. Through further debt-for-equity swaps, the Danish state's investment in Roskilde Bank's equity and subordinated debt reached DKK 12.4 billion at the end of 2010. At the time, its outstanding loss on the restructuring of Roskilde Bank stood at DKK 8.9 billion, not including interest and commissions earned and any later gains on asset sales. It is unclear whether the state's remaining loans and equity in the bank were ever repaid or whether FSC has remaining legacy assets from Roskilde.

Keywords: Denmark, deposit guarantee, restructuring, Roskilde Bank

¹ This case study is part of a Yale Program on Financial Stability (YPFS) selection of New Bagehot Project modules considering bank resolutions and restructurings. A survey of all the cases in this series (McNamara et al. 2024) and the individual cases underlying it are available from the *Journal of Financial Crises* at <https://elischolar.library.yale.edu/journal-of-financial-crises/vol6/iss1/>.

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Overview

This module describes the restructuring and sale of Roskilde Bank A/S (Esmerk 2008f). For more on emergency liquidity to Roskilde, see Decker (forthcoming).

Roskilde was the eighth-largest bank in Denmark, with DKK 43 billion (USD 9.1 billion)³ in consolidated assets as of March 2008 (NAO 2009).

Roskilde had considerable exposure to real estate and construction firms in summer 2008 (AFP 2008a; DFSA 2009; Ministry of Business and Growth 2013; Sjögren and Iversen 2019). Real estate assets comprised more than 43% of Roskilde's loan and guarantee portfolio at the time, which was the main contributor to Roskilde's failure since it had almost no US subprime asset exposure and ample access to central bank funds (AFP 2008a; FT 2008; Ministry of Business and Growth 2013). Throughout the mid-2000s, Roskilde was one of the two fastest-growing regional, midsized banks amid an expanding credit culture (see Figure 1) (DFSA 2009; Ministry of Business and Growth 2013; Sjögren and Iversen 2019). The US housing market crisis and subsequent credit crunch caused Roskilde to take larger write-downs than expected (DNB 2009; NAO 2009).

On July 1, 2008, Moody's Investors Service downgraded Roskilde (see Figure 2) (Moody's 2008a). Roskilde later said that it explored the possibility of raising additional capital after the Moody's downgrade but concluded that such a capital increase could not be underwritten⁴ (Dow Jones 2008a; EC 2008a). Roskilde's board said it feared a

Key Terms

Purpose: When a private buyer did not emerge, DNB and PCA took over Roskilde Bank to avoid bankruptcy, to protect financial stability, and to maintain Danish banks' ability to fund themselves in international markets

Size and Nature of Institution Eighth-largest bank in Denmark, with DKK 43 billion in consolidated assets at end-March 2008

Source of Failure High exposure to real estate and weakening asset quality prompted ratings downgrades

Start Date August 24, 2008

End Date Final winding up ongoing

Approach to Resolution and Restructuring DNB and PCA established a new bank (New Roskilde) to take over all assets and senior liabilities; three banks bought 21 of Roskilde's branches from New Roskilde; FSC took over New Roskilde in August 2009

Outcomes PCA wrote off DKK 750 million in 2009; state losses were DKK 8.9 billion as of the end of 2010; original depositors and senior creditors were made whole

Notable Features PCA participation in the intervention and taking the first loss; earn-out mechanism for existing holders of subordinated debt and equity; ability for buyers of assets from New Roskilde to return deposits and loans; FSC settled with the Consumer Ombudsman on the sale of bank shares by Old Roskilde and offered 17,000 private customers compensation equal to 60% of their net losses

³ Per OFX, USD 1 = DKK 4.72 on March 31, 2008.

⁴ By July 11, 2008, Roskilde's shares had fallen by 78% since the beginning of the year—six times as much as the OMX Copenhagen 20 Index, which was down by 13% (EC 2008a).

bank run and thus sought assistance from Danmarks Nationalbank (DNB) (AFP 2008a; DNB 2009; EC 2008a).

Figure 1: Annual Growth of Total Credit in Large Commercial Banks (Percentage)

	2003	2004	2005	2006	2007
Roskilde	13	39	55	59	37
Danske Bank	—	11	20	15	25
Nordea	0	10	17	-3	14
Jyske Bank	-33	17	22	18	25
Sydbank	14	22	28	20	3
Amagerbanken	-4	16	59	38	36

Source: Sjögren and Iversen 2019.

From July 7–10, 2008, Roskilde’s management met with the Danish Financial Supervisory Association (DFSA) (DFSA 2009; DNB 2008a; Roskilde Bank 2008d). On July 10, 2008, DNB announced it would “provide the necessary liquidity” to Roskilde (DFSA 2009; DNB 2008a). DNB later confirmed this was essentially an unlimited credit facility (EC 2008c; DNB 2023; Dow Jones 2008a). Also on July 10, DNB said that the government and private sector had agreed to cover any losses DNB might bear from the facility (DNB 2008a). The government later said that the Danish Private Contingency Association (PCA) had agreed to guarantee the first DKK 750 million of any potential losses on the liquidity facility, and the government would absorb losses beyond the initial DKK 750 million (EC 2008a; NAO 2009; DFSA 2009). The agreement between DNB and Roskilde contained multiple conditions, including the full or partial sale of Roskilde (AFP 2008a; DNB 2009). On July 10, 2008, Roskilde put itself up for sale (Dow Jones 2008a; Reuters 2008).

On August 24, 2008, DNB announced that it and the PCA acquired Roskilde to avoid liquidation, as Roskilde had not received any private offers (DNB 2008b; DNB 2009; McLaughlin and Levring 2008). The PCA converted its guarantee into an equity stake of DKK 750 million (EC 2008a; EC 2011). To effectively accomplish the takeover of Roskilde, DNB and the PCA established an entity called New Roskilde to which they transferred all assets, depositors, and other senior liabilities of “Old Roskilde.” To ensure that New Roskilde was adequately capitalized, DNB and the PCA made a capital injection of DKK 4.5 billion into New Roskilde. Subordinated liabilities and equity remained behind in “Old Roskilde,” which began insolvency proceedings (EC 2008b).

In September 2008, three banks agreed to buy 21 of Roskilde’s branches (Dow Jones 2008c). The three banks took over DKK 9.3 billion in loans and DKK 4.9 billion in deposits in total. However, the three banks returned loans and deposits that they did not want to take over, at book value (see Key Design Decision No. 10, Treatment of Clients) (EC 2011; Roskilde Bank 2009). This prolonged the Financial Stability Company’s (FSC’s) resolution of Roskilde longer than expected (EC 2011).

In October 2008, the Danish Parliament created the FSC as a more permanent means of resolving banks (Danish Parliament 2008b, pt. 2; NAO 2009; OECD 2009). In August 2009,

the FSC took over New Roskilde (Bjerre-Nielsen 2011; IMF 2014). The resolution process took longer than Danish authorities expected when the banks that bought Roskilde's branches returned a large number of loans and accompanying deposits (EC 2011). In 2011, Roskilde became part of FS Finans, a newly created subsidiary of the FSC for resolving loans and advances (FSC 2011b; FSC 2012).

At the end of 2010, the Danish state's outstanding loss on the restructuring of Roskilde Bank stood at DKK 8.9 billion, not including interest and commissions earned and any later gains on asset sales (FSC 2011b). By September 2014, the FSC no longer had customer liabilities from Roskilde (FSC 2015). It is unclear whether the state's remaining loans and equity in the bank were ever repaid or whether FSC has remaining legacy assets from Roskilde. The FSC reported positive earnings throughout the 2010s based on interest and commission income from member banks (MIBFA n.d.).

In December 2013, the FSC settled with the Consumer Ombudsman on the sale of bank shares by Old Roskilde to 17,000 private customers; the FSC offered these customers compensation equivalent to 60% of their net loss (see Key Design Decision No. 9, Treatment of Creditors and Equity Holders) (Danish Consumer Ombudsman 2013; FSC 2014c).

Figure 2: Timeline of Roskilde's Restructuring and Sale

July 1, 2008	Moody's downgrades Roskilde because of the bank's exposure to real estate and weakening asset quality.
July 10, 2008	Roskilde announces extraordinary losses and puts itself up for sale per agreement with the government. Roskilde's shares fall 57% after the announcement. DNB announces emergency liquidity assistance in the form of an unlimited credit facility. Denmark's Treasury guaranteed any losses that DNB might suffer related to the emergency liquidity facility provided to Roskilde.
July 15, 2008	Moody's further downgrades Roskilde's BFSR, subordinated debt, and preferred stock.
Aug. 24, 2008	DNB and the PCA acquire Roskilde, transferring its assets and senior liabilities to a newly created entity called "New Roskilde" and leaving subordinated liabilities and equity behind in "Old Roskilde," which enters bankruptcy proceedings.
Sep. 25, 2008	Danish authorities hold meetings with four potential buyers of New Roskilde assets. One buyer withdraws. The other three buyers begin negotiating specific contractual terms.
Sep. 29, 2008	Nordea, Spar Nord, and Arbejdernes Landsbank agree to buy 21 of New Roskilde's branches (see Key Design Decision No. 8, Approach to Resolution/Restructuring).
Oct. 6, 2008	Danish authorities approve the transfer of assets and senior liabilities from Old Roskilde to New Roskilde, retroactive as of August 24, 2008. The PCA and DNB make a capital injection of DKK 4.5 billion into New Roskilde.
Jan. 30, 2009	DNB converts DKK 4 billion of its loans to Roskilde Bank and DKK 1 billion of its subordinated loans to the bank into equity to recapitalize Roskilde Bank.

Date	Event
Aug. 10, 2009	The FSC takes control of New Roskilde from DNB for winding up; PCA's stake is written off.
Dec. 30, 2009	FSC converts DKK 2 billion of its outstanding loans to the bank into equity to recapitalize Roskilde Bank.
June 30, 2010	FSC converts DKK 2.7 billion of its outstanding loans to the bank into equity to recapitalize Roskilde Bank.
Jan. 27, 2011	By January 27, the FSC submits an updated liquidation plan for New Roskilde to the European Commission.
June 2011	FSC restructures Roskilde assets and liabilities into FS Finans, a subsidiary of the FSC.
2012	The state reduces its obligation to the FSC to cover Roskilde Bank losses by DKK 4.6 billion in return for further capital distributions.
Dec. 2013	The FSC settles with the Consumer Ombudsman on the sale of Old Roskilde's shares to private customers.
2015	The state reduces its obligation by DKK 1.8 billion in return for further capital distributions.
2018	The state reduces its obligation by DKK 2.5 billion in return for further capital distributions.

Source: Author's analysis.

Summary Evaluation

On September 18, 2008, the Public Accounts Committee—appointed by parliament—asked the National Audit Office (NAO) to review the role of the DFSA in Roskilde's collapse and resolution (see Key Design Decision No. 5, Governance) (Esmerk 2008d; NAO 2009).

In June 2009, the NAO returned a report with findings to the Public Accounts Committee. With regard to the bank's collapse, the NAO found that the DFSA supervised Roskilde according to relevant guidelines but failed to conduct a sufficiently systemic follow-up—including fully maintained estimates of the bank's capital adequacy need after the DFSA became aware of Roskilde's problems (NAO 2009). See Figure 3 for Roskilde's capital adequacy ratio (CAR) for the years preceding insolvency.

Figure 3: Roskilde's Capital Adequacy Ratio (CAR)

	2003	2004	2005	2006	2007
CAR	12.9%	12.2%	11.5%	12.8%	11.9%
Core Capital Ratio	8.6%	8.6%	8.3%	7.9%	7.0%

Note: The capital adequacy need represented the credit institution's assessment of the minimum capital adequacy ratio needed by the bank. The CAR was the base capital as a percentage of risk-weighted items. In 2007, Roskilde had a required CAR of 10.5%.

Sources: NAO 2009; Roskilde Bank 2008c.

With regard to the resolution of Roskilde, the NAO wrote the topline conclusion:

The [liquidity facility and takeover of] Roskilde Bank A/S were developed ad hoc, implemented on the basis of a number of assumptions made by the Ministry of Economic and Business Affairs and established under extreme time constraints. In the course of the investigation, the Ministry of Economic and Business Affairs procured the necessary statutory authority to leave the actual winding up of Roskilde Bank A/S to the central government winding up company (the Financial Stability Company). (NAO 2009, 21)

On June 17, 2009, the DFSA released its own report on the causes of Roskilde's collapse and with a self-assessment of the DFSA's response. The DFSA said that it supervised Roskilde within the framework of contemporaneous laws and did not act beyond that authority. The DFSA partly attributed the failure of the private acquisition of Roskilde in July–August 2008 to the takeover of bankTrelleborg by Sydbank A/S in June 2008. As in the case of bankTrelleborg, Roskilde had a large deposit deficit that would have to be refinanced by prospective buyers after a merger. It would have been difficult to obtain even short-term financing during that time, according to the DFSA. The DFSA also noted that a significant part of Roskilde's capital base came from the sale of shares to customers (DFSA 2009).

At the conclusion of the International Monetary Fund's (IMF's) mission to Denmark on October 2, 2008, the IMF said: "The authorities' preemptive approach to bank resolution has worked well so far" (IMF 2008a, 2). Earlier in 2007, the Danish Bankers Association (DBA) established the PCA with the ability to grant up to DKK 750 million in guarantees (IMF 2008a; PCA 2011).⁵

In an Article IV staff report published in December 2008, the IMF said: "The proactive and collaborative approach to bank resolution has helped prevent contagion during a period of exceptionally fragile confidence" (IMF 2008b, PDF p. 36). The IMF praised the burden-sharing with the PCA—this private sector participation incentivized banks to seek private sector solutions since shareholders faced resolution. Burden-sharing furthermore protected the taxpayer from the bank's losses. Some of the features in Roskilde's resolution were incorporated into Denmark's new Financial Stability Act, whose framework the IMF characterized as "innovative and comprehensive." These features included: (1) guarantee of

⁵ The author cites the PCA's articles of association from 2011 to discuss its administration because research during the writing of this case did not uncover an earlier version.

claims, (2) resolution body, (3) burden sharing, (4) level playing field, (5) addressing moral hazard and intensifying supervision (IMF 2008b).

In a 2013 article published in *New Political Economy*, Martin B. Carstensen notes Roskilde's role in the evolution of Danish bank resolution. Before Roskilde's collapse, Danish authorities would offer guarantees to specific institutions and allow other healthier banks to take over distressed banks. This was consistent with a strict non-state ownership policy, according to Carstensen. He points to the example of Sydbank's acquisition of bankTrelleborg in June 2008. Roskilde, as the eighth-largest lender in Denmark, could not be sold during the midst of the financial crisis, so it was resolved with a state guarantee that was, as Carstensen argues, "very expensive for the state" (Carstensen 2013, 568).

Context: Denmark, Roskilde, 2007–2008	
Assets	DKK 42.0 billion as of Dec. 31, 2007 DKK 37.3 billion as of Aug. 29, 2008
Liabilities	DKK 42.0 billion as of Dec. 31, 2007 DKK 37.3 billion as of Aug. 29, 2008
Deposits	DKK 17.4 billion as of Dec. 31, 2007 DKK 17.7 billion as of Aug. 29, 2008
Capital Ratio	Core capital ratio: 7.0% as of Dec. 31, 2007 Core capital ratio: “negative” as Aug. 29, 2008
Nonperforming Loans	Not available Not available
Market Share	Eighth-largest bank in Denmark as of March 31, 2008
Banking System, % of GDP	205.6% in 2007 219.1% in 2008
<i>Sources: NAO 2009; Roskilde Bank 2008c; Roskilde Bank 2008d; World Bank.</i>	

Key Design Decisions

- 1. Purpose: When a private buyer did not emerge, DNB and the PCA acquired Roskilde to avoid bankruptcy, to protect financial stability, and to maintain Danish banks' ability to fund themselves in international markets.**

Roskilde had considerable exposure to real estate and construction markets in July 2008. Rating agencies downgraded Roskilde when the bank's management discovered that Roskilde would have to take larger write-downs than expected (AFP 2008a; DNB 2009; EC 2008a; NAO 2009; Dow Jones 2008a). Roskilde's board said later that it feared a bank run and thus sought assistance from DNB. DNB and the PCA agreed to grant an unlimited credit facility, conditioned on Roskilde's full or partial sale to simultaneously avoid speculation about suspension of payments and avoid setting the precedent of a governmental safety net for other distressed banks (AFP 2008a; DNB 2009; EC 2008a; NAO 2009). On July 10, 2008, Roskilde put itself up for sale (Dow Jones 2008a; Reuters 2008). The Danish government and DNB chose PricewaterhouseCoopers (PwC) to verify and approve the cashflow budget required for each drawdown of the credit facility for the next 14 days (EC 2008a).

On August 24, 2008, since the bank did not receive any offers after putting itself up for sale, DNB and the PCA acquired Roskilde to avoid liquidation (DNB 2009; McLaughlin and Levring 2008). DNB and the Ministry of Economic and Business Affairs said that they wanted to avoid bankruptcy proceedings for Roskilde in order to protect financial stability and Danish banks' ability to fund themselves in international markets (EC 2008b; NAO 2009; McLaughlin and Levring 2008). On the choice of a bridge bank, then-DNB Governor Nils Bernstein said, "The purpose of the new bank is to continue banking business in order to maximize the proceeds from the discontinuation of the business taken over from Roskilde Bank" (Bernstein 2008). Danish authorities chose to create New Roskilde because taking assets and liabilities from Old Roskilde directly would have entailed expropriation and a complicated legal process that Danish authorities did not deem appropriate for Roskilde's situation. Through New Roskilde, Danish authorities could pursue a progressive sale or liquidation and ensure that all of Roskilde's senior creditors were repaid. Danish authorities avoided a "difficult" valuation of New Roskilde through the creation of an earn-out mechanism (see Key Design Decision No. 9, Treatment of Creditors and Equity Holders). During the sales process of New Roskilde's branches, Danske Markets served as a financial advisor and issued a fairness opinion: the sales prices achieved were the maximum market prices, and there was no advantage to the acquirers of the branches (EC 2008b).

In 2008, DNB and the PCA made a capital injection of DKK 4.5 billion into New Roskilde to meet regulatory solvency requirements and to protect existing depositors and other senior creditors from losses (EC 2008b; IMF 2014).

In 2011, Roskilde became part of FS Finans, a newly created subsidiary of the FSC for winding up loans and advances (FSC 2011b; FSC 2012).

2. Part of a Package: DNB and the PCA granted Roskilde an unlimited credit facility while the bank put itself up for sale.

Unlimited Credit Facility (Emergency Liquidity Assistance)

After discussion with the Ministry of Economic and Business Affairs, the DFSA, and the DBA, DNB agreed to grant emergency liquidity assistance in the form of an unlimited credit facility on July 10, 2008 (EC 2008c; ENP 2008; NAO 2009). The credit facility was unlimited in duration and amount. The PCA guaranteed the first DKK 750 million of any potential losses on the credit facility, and DNB absorbed losses beyond the initial DKK 750 million, backed by a guarantee from the Danish state that the Danish Parliament approved in July 2008 (EC 2008a; NAO 2009). The PCA did not receive a fee for the guarantee (EC 2008a). The PCA's guarantee was an indemnity bond, which meant that the PCA paid under the guarantee as soon as Roskilde failed to repay its loan to DNB, regardless of whether Roskilde entered insolvency proceedings. The interest rate for the credit facility was set at 2 percentage points above DNB's lending rate; DNB's lending rate at the time was 4.6%, so the credit facility's interest rate was set at 6.6% (EC 2008a).

Roskilde could use the credit facility only to cover liquidity that was necessary for the bank's day-to-day activities. Roskilde could not use the facility to invest in new activities; take on risky positions in commercial markets; engage in new, more extensive credit facilities; or actively market and attract new depositors or borrowers. DNB and PricewaterhouseCoopers verified the cashflow budget for the next 14 days with each drawdown of the facility. PwC particularly scrutinized whether the budgeted need complied with the behavioral conditions attached to the credit facility (for example, no investments in new activities) (EC 2008a).

On July 18, 2008, the minister for economic and business affairs told the Finance Committee of the Danish Parliament that he did not expect drawdowns on the unlimited credit facility to exceed DKK 15 billion to DKK 20 billion (EC 2008a). However, after Lehman Brothers failed and led to market stress across the world, the European Commission (EC) reported that the bank drew down at least DKK 36.8 billion on the facility by early October on a short-term basis to allow the bank to pay depositors and other senior creditors by the end of that month (EC 2008b). Roskilde had DKK 17.5 billion outstanding on the facility at the end of 2008, according to DNB's annual report for 2008 (DNB 2009).

For more on the unlimited credit facility, see Decker (forthcoming).

State Guarantee

On July 11, 2008, the Treasury guaranteed any losses that DNB might suffer related to the emergency liquidity facility provided to Roskilde. The guarantee was provided for an unlimited amount without a final maturity date and without a fee. The Treasury needed approval from the Finance Committee in parliament for the guarantee (see Key Design Decision No. 3, Legal Authority) (EC 2008a). For more on Denmark's General Guarantee Scheme during the Global Financial Crisis (GFC) of 2009–2009, see Sabath (2020).

Deposit Guarantee

Effective October 10, 2008, Minister for Economic and Business Affairs Lene Espersen announced a two-year General Guarantee Scheme for all bank deposits,⁶ excluding deposits at Roskilde (FSC 2011a; FSC 2011b; Levring 2008). However, an agreement between the FSC and the PCA signed on August 10, 2009, provided DKK 10.5 billion from the General Guarantee Scheme to cover accounts at Roskilde (FSC 2011a; FSC 2011b).

3. Legal Authority: The Financial Business Act allowed DNB to take over Roskilde; the Financial Stability Act allowed the FSC to complete Roskilde's resolution.

The DBA established the PCA on June 13, 2007, to help distressed banks (EC 2008a; PCA 2011). Per section 3 of its articles of association, the PCA can provide guarantees during the resolution of a distressed bank by the FSC; however, the PCA's guaranteed amount cannot exceed the government's estimated loss if the distressed institution undergoes bankruptcy proceedings (PCA 2011). The PCA cannot itself take over a distressed bank (PCA 2011).

No legal and organizational framework existed on the orderly resolution of banks before the GFC (DNB 2023). However, Danish authorities made use of existing powers from various legislation to resolve Roskilde while concurrently undergoing financial sector reform.

At the time of Roskilde Bank's failure, Article 246 of Denmark's existing Financial Business Act allowed the transfer of Roskilde's activities to the government because the bank did not meet solvency requirements; this transfer also required the approval of the minister for economy and business (Danish Parliament 2008a, arts. 124, 246; EC 2008b).

The Financial Business Act also allowed the DFSA to supervise foreign banking institutions, including Nordea (Danish Parliament 2008a, secs. 243, 344; IMF 2007). Nordea's acquisition of Roskilde's branches was subject to approval by the DFSA (IMF 2007).

The Finance Committee of the Danish Parliament passed Appropriation 199 4/9 2008 concerning the state guarantee for any loss suffered by DNB for the unlimited credit facility. It was this commitment that led to the government's DKK 8.9 billion loss (EC 2008a; FSC 2011b; NAO 2009).

On October 10, 2008, the Danish Parliament passed the Act on Financial Stability (or Financial Stability Act) to establish the General Guarantee Scheme for deposits and other senior liabilities of banks that were members of the PCA, which did not include Roskilde⁷ (FSC 2010; DMEA and DMoF 2008).

⁶The General Guarantee supplemented an existing, privately funded deposit insurance system administered by DNB, which covered deposits of all PCA members up to EUR 50,000 (DGFDI 2009; PCA 2011). For more on Denmark's General Guarantee Scheme, see Hoffner (2022).

⁷ According to the EC, Roskilde was a member of the PCA at its founding on June 13, 2007 (EC 2008a). However, primary source documents from the FSC indicate that Roskilde was not a member of the PCA at the time of the bank's resolution (FSC 2010).

On July 31, 2008, the EC ruled that the part of the unlimited credit facility guaranteed by the PCA did not constitute State Aid, while the part of the unlimited credit facility guaranteed by DNB did constitute State Aid compatible under the application of Article 87(3)(c) EC Treaty and the Community guidelines on State Aid for rescuing and restructuring firms in difficulty (see Figure 4) (EC 2008b; EC 2008c).

On August 24, 2008, Danish authorities initially decided to acquire Roskilde through DNB instead of the FSC because the Ministry of Economic and Business Affairs expressed worry that use of the FSC would complicate its pending approval by the EC. The FSC was officially established on October 13, 2008 (NAO 2009).

On May 24, 2011, the EC found Denmark's amended liquidation aid measures to be compatible with the internal market under Article 107(3)(b) of the Treaty on the Functioning of the European Union (TFEU). The EC said that the amended measures closely resembled the initial liquidation aid measures; the amended measures also served the same purpose of allowing a solvent liquidation of Roskilde (EC 2011).

Figure 4: Timeline of EC Decision-Making

Date	Event
July 15–24, 2008	Danish authorities submit the details of the unlimited credit facility to the EC.
July 31, 2008	The EC authorizes aid measures pertaining to the unlimited credit facility.
Sep. 16, 2008	Danish authorities send a draft notification of the measures for Roskilde.
Sep. 23, 2008	Danish authorities meet with EC representatives.
Sep. 25, 2008	Danish authorities contact the EC to inform the regulatory body about the sales process.
Sep. 26, 2008	Danish authorities provide a summary of the intended deal.
Oct. 7, 2008	The EC receives a formal notification of the aid measures.
Nov. 5, 2008	The EC does not object to Danish authorities' aid measures.
May 24, 2011	The EC rules that the amended liquidation measures are compatible with the internal market under Article 107(3)(b) of the TFEU.

Sources: EC 2008b; EC 2011.

4. Administration: Roskilde's guarantee, takeover, and sale were negotiated among the Ministry of Economic and Business Affairs, DNB, and PCA. DNB was responsible for the administration of Roskilde Bank itself until the FSC took over in 2009.

DNB and the PCA took over Roskilde between its failure to initiate a sale in August 2008 and August 2009, when the FSC took over the remainder of Roskilde. Although the FSC was created in October 2008, the Ministry of Economic and Business Affairs believed that using the FSC to assume ownership of Roskilde's assets would complicate the EC's pending approval. In June 2011, the FSC reorganized the assets and liabilities from various banks under a new group structure.

DNB and the PCA

Roskilde's board said it feared a bank run and thus sought assistance from DNB on July 10, 2008 (AFP 2008a; DNB 2009; EC 2008a; Roskilde Bank 2008d). DNB consulted the Ministry of Economic and Business Affairs, DFSA, and DBA before granting emergency liquidity assistance in the form of an unlimited credit facility—of which the PCA guaranteed the first DKK 750 million—while Roskilde put itself up for sale (AFP 2008a; DNB 2009; ENP 2008; EC 2008c; NAO 2009).

On July 15, 2008, Roskilde hired Danske Markets as its investment banker to market the company to prospective buyers (Bernstein 2008; Roskilde Bank 2008d).

DNB was responsible for Roskilde's administration and related costs after the government took over the distressed institution on August 24, 2008; the PCA itself could not take over a distressed bank, according to the PCA's articles of association (NAO 2009; PCA 2011). DNB received weekly summaries of its account with Roskilde. DNB regularly notified the minister for economic and business affairs and the Finance Committee of the Danish Parliament about losses absorbed by the government (NAO 2009).

According to the NAO report, the Ministry of Economic and Business Affairs did not transfer Old Roskilde to the FSC during autumn 2008 because such a transfer would have complicated the EC's pending approval of the not-yet-established FSC at the time (NAO 2009).

Soon after DNB took over Roskilde, current and former employees from Nykredit, Nordea, and Danske Bank helped review Roskilde's existing credit engagements (Roskilde Bank 2009).

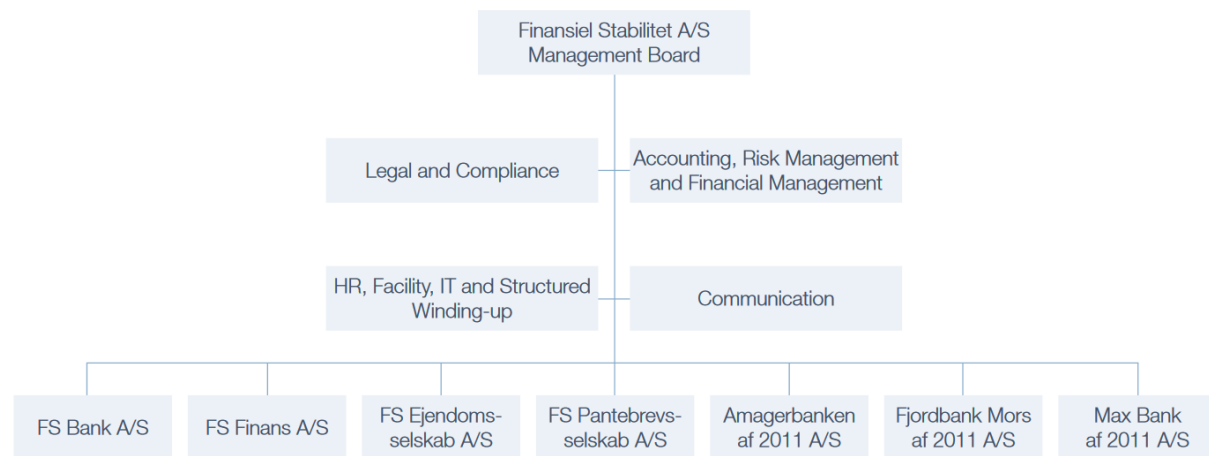
FSC

On August 10, 2009, the FSC took over New Roskilde for resolution (Bjerre-Nielsen 2011; IMF 2014). The Danish state owns the FSC through the Ministry of Economic and Business Affairs (NAO 2009). New Roskilde maintained a security dealer license for legal purposes, but the bank's functions were limited (EC 2011).

The management board, comprising one member, was responsible for the day-to-day administration of the FSC. The FSC's board of directors appoints the management board (FSC 2011b).

FS Finans and FS Bank

In June 2011, the FSC reorganized its group structure to ensure efficiency, coordination, and quality in resolution processes, considering that the government had taken over a number of banks (see Figure 5)(FSC 2011b; FSC 2012). The FSC also said that this reorganization would ensure uniform, continuous, and fair treatment of customers of banks taken over (FSC 2011b).

Figure 5: FSC 2011 Reorganization

Source: FSC 2012.

Under the new structure, FS Finans A/S became the resolution company for the majority of the FSC's loans and advances, including Roskilde's (FSC 2011b; FSC 2012).

Any remaining corporate customers of New Roskilde were transferred to FS Bank A/S (see Key Design Decision No. 10, Treatment of Clients) (FSC 2011b). FS Bank was the subsidiary focused on helping customers find another private bank; FS Bank's customer portfolio mainly comprised corporate customers. Thus, FS Bank offered limited services to corporate customers, such as credit lines, debit cards, and online banking. FS Bank was in charge of payment transfers and acted as a custodian bank for all financial charges for FS Finans among others (FSC 2012).

5. Governance: The government replaced Roskilde's management.

Investigating Roskilde's Supervision and Collapse

On September 18, 2008, the Public Accounts Committee asked the NAO⁸ to review the role of the DFSA in Roskilde's collapse and resolution (Esmerk 2008d; NAO 2009). The NAO submitted its report on the DFSA's role in the collapse and winding up of Roskilde in accordance with the Auditor General's Act, Sections 8(1) and 17(2), amended by Act No. 590 of June 13, 2006. In June 2009, the NAO returned a report with findings to the Public Accounts Committee (see Summary Evaluation) (NAO 2009).

⁸ The NAO is an independent institution placed under parliament and tasked with investigating questions of public finance, compliance, and performance (NAO n.d.a). The Public Accounts Committee—which verifies whether government accounts are correct, assesses whether public funds are managed effectively, and submits audited government accounts to parliament for approval—may also ask the NAO to investigate certain matters (NAO n.d.b; Public Accounts Committee of the Danish Parliament 2012).

The DFSA had to submit a report on the causes of Roskilde's failure and the DFSA's role in the process leading up to bankruptcy; the report was released to public on June 17, 2009 (DFSA 2009; Ministry of Business and Growth 2013).

The FSC launched an external legal investigation into the causes of Roskilde's collapse and whether there were grounds for lawsuits against the bank's former management (Ministry of Business and Growth 2013).

Roskilde's Governance under DNB

DNB and the PCA appointed new board members on August 29, 2008 (see Key Design Decision No. 12, Treatment of Board and Management). The newly appointed chairman and deputy chairman—Henning Kruse Petersen and Jakob Brogaard, respectively—were also the chairman and deputy chairman of the FSC (DNB 2008c; FSC 2010; FSC 2011b; Roskilde Bank 2009).

On September 12, 2008, a new board of directors was elected to New Roskilde at an extraordinary general meeting (see Key Design Decision No. 12, Treatment of Board and Management) (Esmerk 2008b; Roskilde Bank 2008a; Roskilde Bank 2008b).

Roskilde's Governance under the FSC

The FSC is governed by a board of directors to ensure that it is properly managed as required by law and by the articles of association. Seven members comprise the board, including a chairman and deputy chairman, for terms of three years with the possibility of one reelection (Danish Parliament 2008b, art. 2(6); FSC 2011b). The board is required to notify the minister for economic and business affairs on matters of material financial or political significance relating to the FSC's activities (FSC 2011b).

6. Communication and Disclosure: Senior government officials characterized Roskilde's takeover as necessary given the bank's size, concerns about financial stability, and the need to maintain continued access to foreign financing.

On August 24, 2008, DNB issued a statement describing the Roskilde Bank situation as "very serious. It is expected that the takeover of Roskilde Bank will contribute to limiting the negative effect on the Danish financial system" (DNB 2008b).

On August 27, 2008, then-DNB Governor Nils Bernstein spoke at a press conference about Roskilde. He said that Roskilde had a "lenient credit culture" and was no longer solvent (Bernstein 2008). Bernstein further said that Roskilde's situation posed a significant threat to financial stability in Denmark. He characterized the takeover by DNB and the PCA as a move to protect stability and preserve other Danish banks' access to international capital markets. On the treatment of creditors and equity holders, Bernstein said that shareholders and holders of hybrid core capital and subordinated loan capital would likely lose their capital (see Figure 10) (Bernstein 2008).

Toward the end of September 2008, Economic and Business Affairs Minister Lene Espersen said that solvent banks should cope with the crisis on their own. The government took over Roskilde because of its size and concerns about financial stability, according to Espersen (Esmerk 2008e).

In June 2009, the NAO released a report on the DFSA's role in the collapse and winding up of New Roskilde. The Ministry of Economic and Business Affairs and the FSA allowed the NAO to release the report without redactions (NAO 2009).

On June 17, 2009, the DFSA released its own report on the causes of Roskilde's collapse. The DFSA also self-assessed its response. Usually, such reports could not be made public by the DFSA or minister for economy and business. However, the DFSA was allowed to publicize its findings because the central bank had provided a liquidity facility to Roskilde, guaranteed by the government, and Roskilde had been declared bankrupt (DFSA 2009).

7. Source and Size of Funding: The Danish government funded liquidity support, restructuring, and resolution through appropriations passed by the Finance Committee in the Danish Parliament.

DNB and PCA Takeover

The Finance Committee of the Danish Parliament passed Appropriation 199 4/9 2008 concerning the state guarantee for any loss suffered by DNB for the unlimited credit facility. The Finance Committee initially expected the drawdowns on this facility to total DKK 15 billion to DKK 20 billion, though the EC reported that usage was over DKK 36.8 billion by early October 2008 (see Key Design Decision No. 2, Part of a Package) (EC 2008b; NAO 2009). Roskilde had DKK 17.5 billion outstanding on the facility at the end of 2008, according to DNB's annual report for 2008 (DNB 2009).

Private sector support from the PCA reflected negotiations among DNB, the Ministry of Economic and Business Affairs, and the PCA. While no calculations were performed for private sector support, the PCA's contribution could not exceed the government's estimated loss if Roskilde were to enter bankruptcy, per the PCA's articles of association (NAO 2009; PCA 2011). The PCA was funded by voluntary contributions from member banks and from Nykredit (NAO 2009; PCA 2011). Member banks contributed DKK 750 million per year, the maximum annual contribution, unless otherwise unanimously decided by the board of representatives.⁹ The PCA guaranteed a maximum of DKK 1.5 billion per calendar year, though the board of representatives could change this amount with a unanimous vote (EC 2008a; NAO 2009; PCA 2011). The PCA wrote off its DKK 750 million investment in Roskilde after the government took over (FSC 2010).

On August 29, 2008, the chairman of the Finance Committee received a letter from the Ministry of Economic and Business Affairs about additional appropriations needed. Days earlier, it became apparent that Roskilde had to make further write-downs that would prevent the bank from meeting the calculated capital adequacy need and capital requirement of the Financial

⁹ Per the World Bank, the size of Denmark's gross domestic product was USD 353.4 billion in 2008.

Services Act.¹⁰ Because the EC had already approved the measures necessary for the emergency liquidity assistance, and because these measures had already been taken, the Ministry thought that Appropriation 199 could not be adjusted. Thus, the Finance Committee had to pass additional appropriations. On November 4, 2008, the Finance Committee passed Appropriation 25 27/11 2008 to extend the government's guarantee to include any losses incurred in connection with DNB's takeover and resolution of Roskilde (NAO 2009).

The Act on Financial Stability in October 2008 (otherwise known as the Bank Package I) did not cover Roskilde's restructuring and resolution (FSC 2010; FSC 2011b).

FSC Takeover

On August 10, 2009, DNB and the PCA transferred all shares in Roskilde to the FSC. DNB also transferred a subordinated loan to the FSC. The FSC took over liabilities related to DNB's emergency liquidity assistance to Roskilde. Because Roskilde was not a member of the PCA, the FSC accounted for its costs separately from the FSC's other activities under the Financial Stability Act (FSC 2010).

At the end of 2010, retained losses—representing the loss to the Danish State—reached DKK 8.9 billion, not including interest and commissions earned and any later gains on asset sales. The FSC reported that amount as an asset on its balance sheet, representing a receivable due from the Danish State (FSC 2011b). It is unclear whether the State's remaining loans and equity in the bank were ever repaid. The FSC reported positive earnings throughout the 2010s based on interest and commission income from member banks (MIBFA n.d.).

At the termination of Denmark's guarantee scheme on September 30, 2010, the FSC would appoint auditors to calculate profits and losses: equity at the end of the period minus capital contributions in subsidiaries plus market return (Danish Parliament 2008b, sec. 9.4, 17.4). If the FSC's financial statements showed a loss, the PCA guaranteed up to DKK 10 billion. If losses exceeded that amount, the PCA's annual guarantee commission of DKK 7.5 billion would be used. If the FSC's total losses exceeded these amounts, the PCA would further guarantee additional losses up to DKK 10 billion (Danish Parliament 2008b, sec. 9.5).

8. Approach to Resolution and Restructuring: Roskilde Bank's restructuring and resolution involved the creation of a new bank to take over all assets and senior liabilities of Roskilde, partial sale to three banks, and winding up by the FSC.

Acquisition by the Government

Old Roskilde initially intended to sell itself during a window from July 15 to August 22, 2008 (EC 2008b). After Old Roskilde failed to receive any offers by the expiration date, DNB and the PCA took over the distressed institution. The Ministry of Economic and Business Affairs

¹⁰ The capital adequacy requirement was determined by the following: the base capital had to amount to 8% or more of the institution's risk-weighted items or capital adequacy need if this was higher than 8% (NAO 2009).

and DNB said that they did not want Roskilde to declare bankruptcy, which in their opinion would seriously damage financial stability (NAO 2009).

Between August and November 2008, the acquisition of Roskilde by DNB and the PCA involved three measures. First, on August 24, DNB and the PCA signed an agreement to create a bank joint stock company: New Roskilde. Second, on August 24, Roskilde, DNB, and the PCA signed an agreement to transfer all assets and senior liabilities of Roskilde to New Roskilde at a value of DKK 37.3 billion (EC 2008b; NAO 2009). Third, on November 4, 2008, the Finance Committee passed Appropriation 25 27/11 2008 concerning DNB's part in the unlimited credit facility (NAO 2009).

On September 1, 2008, shareholders of Old Roskilde approved the transfer agreement that would allow Danish authorities to move assets, deposits, and senior liabilities to New Roskilde. Danish authorities chose to create the bridge bank, New Roskilde, because taking assets and liabilities from Old Roskilde directly would have entailed expropriation and a complicated legal process that Danish authorities did not deem appropriate for Roskilde's situation. Through New Roskilde, Danish authorities could pursue a progressive sale or liquidation and ensure that all of Roskilde's depositors and senior creditors were repaid (EC 2008b).

Old Roskilde remained operational until October 6, 2008, at which point all assets, deposits, and senior liabilities were transferred to New Roskilde, retroactive as of August 24, 2008. Old Roskilde maintained subordinate liabilities—such as hybrid core capital, subordinated loan capital, the enclosed savings bank trust, and equity—and initiated a proceeding under the Danish Bankruptcy Act (EC 2008b).

Following the acquisition of Roskilde by DNB and the PCA, on October 6, 2008, DNB and the PCA made a capital injection totaling DKK 4.5 billion into Roskilde to meet solvency requirements (EC 2008b; IMF 2014). DNB injected DKK 1.7 billion in the form of class A shares, and the PCA injected DKK 750 million in the form of class B shares. These respective classes of shares ensured that Roskilde's incurred losses would first be covered by the PCA before DNB (EC 2008b; EC 2011).

Under a separate loan agreement, DNB provided DKK 2 billion as Tier 2 capital with a quarterly interest rate set by DNB and an additional risk rate of 4.85% per annum (EC 2008b; EC 2011). Because the loan was Tier 2 capital, it was subordinated relative to other senior debt. However, the loan was senior to share capital. Upon the PCA's capital injection, its previous guarantee of DKK 750 million on July 10, 2008, was released (EC 2008b).

See Figure 6 for the transfer of assets and liabilities out of Old Roskilde.

Figure 6: Pro Forma Transfer of Assets and Liabilities out of Old Roskilde, DKK billions

Old Roskilde (before October 6, 2008)	
Assets	Liabilities & Equity
<u>Total assets:</u> 37.3	<u>Senior liabilities:</u> 37.3
	<u>Subordinated debt:</u> 2.5
	<u>Equity:</u> -2.5
	Left behind

Transferred out

Note: For New Roskilde's balance sheet, see Figure 8.

Source: EC 2008b.

The transfer of senior liabilities to New Roskilde constituted an event of default, enabling creditors to demand redemption of their funds. DNB used its unlimited credit facility to redeem senior liability holders, with the result that as of October 6, 2008, at least DKK 36.8 billion was outstanding under the credit facility (EC 2008b).

For the fourth quarter of 2008, New Roskilde recorded a loss of DKK 3.5 billion due to loan impairments. New Roskilde's equity at the end of 2008 was negative DKK 1.0 billion (DNB 2009; EC 2011). On January 30, 2009, New Roskilde raised its equity capital by DKK 5.0 billion by converting DKK 4.0 billion of DNB's loans to equity and DKK 1.0 billion of Tier 2 subordinated debt to equity. This allowed New Roskilde to bring equity up to just under DKK 4 billion (DNB 2009).

Sale to Nordea, Spar Nord, and Arbejdernes Landsbank

Danish authorities said they sought a quick sale of New Roskilde to maximize the price of assets and liabilities because the value of assets and liabilities tended to decrease with the passage of time since the announcement of Roskilde's wind-up. And indeed, the final sales agreements reflected a decrease in the branch network's value between liquidation announcement and implementation date (see Figure 7) (EC 2008b).

During the second attempt to sell Roskilde, Danish authorities contacted potential buyers that expressed interest during the first sale attempt in addition to new potential buyers. Potential buyers submitted feedback, and Danish authorities used this information to identity prequalified potential buyers. Six prequalified potential buyers received a process letter on the branch network, subject to nondisclosure agreements. Of the six, four prequalified potential buyers agreed to attend a meeting on September 25, 2008, for further discussions. One

prequalified potential buyer opted to withdraw. The other three reached a general agreement with New Roskilde on the division of the branch network and the sale price. The parties initiated a negotiation process for the specific contractual terms (EC 2008b).

On September 29, 2008, three banks agreed to buy 21 of New Roskilde's branches: Nordea, Spar Nord, and Arbejdernes Landsbank (AFP 2008b). Upon sale, the three banks took over, in total, DKK 9.3 billion in loans. They paid DKK 4.9 billion in cash and took over DKK 4.9 billion in deposits, for a total cost of DKK 9.9 billion; the difference was goodwill (EC 2011; IMF 2014).

Figure 7: New Roskilde Bank's Buyers, as of September 2008

Name of Buyer	Branches Acquired	Loans and Guarantees Acquired	Deposits Acquired	Deal Valuation: Announcement	Deal Valuation: Completion
Nordea	9	DKK 5.5 billion	DKK 2.7 billion	DKK 239.6 million	DKK 207.9 million
Spar Nord	7	DKK 2.4 billion	DKK 1.6 billion	DKK 120.0 million	DKK 116.4 million
Arbejdernes Landsbank	5	DKK 1.4 billion	DKK 0.6 billion	N/A	DKK 15.8 million

Note: Values of loans, guarantees, and deposits acquired are prior to returns by the acquiring banks (see Key Design Decision No. 10, Treatment of Clients, and Key Design Decision No. 11, Treatment of Assets).

Sources: AFP 2008b; Capital IQ; EC 2011; IMF 2014; OECD 2009; Nordic News Digest 2008; Spar Nord 2008; Nordea 2008.

Transfer to FSC

New Roskilde's liquidation took longer than Danish authorities expected, as they initially envisioned a two-year process (EC 2008b; EC 2011). At issue was the "large number" of deposits and loans returned by the acquiring banks—more than 50% of corporate commitments were returned (see Key Design Decision No. 10, Treatment of Clients, and Key Design Decision No. 11, Treatment of Assets). Since New Roskilde retained deposits and loans, it could not be liquidated immediately (EC 2011).

On August 10, 2009, the FSC took over New Roskilde for resolution (Bjerre-Nielsen 2011; IMF 2014). In the transfer, the FSC compensated DNB for its outstanding claims on the bank—including DKK 14.5 billion of debt from the credit facility, DKK 1.0 billion of Tier 2 subordinated debt, and DKK 6.7 billion of class A equity shares (FSC 2010). The PCA received no compensation for its shares, per an agreement with the FSC on July 21, 2009 (EC 2011).

As a result of the transfer, the FSC became the 100% shareholder of Roskilde Bank (FSC 2011b). The FSC estimated that the bank had accumulated retained losses of DKK 3.9 billion prior to the transfer, representing the outstanding loss to the Danish state at that time. New Roskilde continued to reevaluate its loan book between August and December 2009, bringing retained losses up to DKK 6.6 billion. In December 2009, to prevent the bank from falling into negative equity, the FSC converted DKK 2 billion of the outstanding balance of the

credit facility into equity (EC 2011; FSC 2010). The FSC converted another DKK 2.7 billion of that debt into equity in June 2010 (EC 2011; FSC 2011b).

At the end of 2010, retained losses—representing the loss to the Danish state—reached DKK 8.9 billion, not including interest and commissions earned and any later gains on asset sales. The FSC reported that amount as an asset on its balance sheet, representing a receivable due from the Danish state (FSC 2011b).

See Figure 8 for changes in New Roskilde's balance sheet from year to year. See Figure 9 for the restructuring of liquidity assistance and capital.

Figure 8: Changes in New Roskilde's Balance Sheet, DKK Millions

	Dec. 31, 2008	Jan. 30, 2009	Aug. 10, 2009	Dec. 31, 2009	Dec. 31, 2010
Assets					
Loans			14,523	12,602	6,541
Total Assets	29,366		18,094	16,261	8,806
Liabilities					
Owed to DNB	17,517	13,517	14,355*		
Owed to Banks				246	406
Owed to FSC				10,578	3,383
Other Liabilities			2,834	2,330	1,431
Deposits			1,577	1,353	658
Other Liabilities	10,877		316	314	317
Provisions			941	663	456
Subordinated Debt	2,000	1,000	1,000	1,000	1,000
Total Liabilities	30,394		18,189	14,157	6,220
Equity					
Equity			(95)	2,104	2,586
Class A	(1,028)	3,972	6,723	8,723	11,423
Class B	1,723	6,723			
Other/Error	750	750		(15)*	94*
Term			(6,818)	(6,604)	(8,931)
Retained Losses	(3,501)	(3,501)	(6,818)	(6,604)	(2,327)
Change					
Total Liabilities and Equity	29,366		18,094	16,261	8,806

Note: Starred figures are inferred calculations.

Sources: DNB 2009; DNB 2010; EC 2011; FSC 2010; FSC 2011b; FSC 2012.

Figure 9: Outstanding Liquidity Assistance to Roskilde and Equity at Subscription Value, in Order of Priority of Claims, as of December 31, 2010

Type	Initial Amount	Restructured Amount
Contingent Amount Available under DNB Credit Facility	DKK 36.8 billion (base rate + 2.0% per annum)	DKK 28.1 billion (base rate + 2.0% per annum)
Amount Outstanding under DNB Credit Facility	DKK 36.8 billion (base rate + 2.0% per annum)	DKK 3.4 billion (base rate + 2.0% per annum)
Tier 2 Capital (from DNB)	DKK 2.0 billion (base rate + 4.9% per annum)	DKK 1.0 billion (base rate + 4.9% per annum)
A-Class Tier 1 Capital (from DNB)	DKK 1.7 billion (base rate + 4.9% per annum)	DKK 11.4 billion (base rate + 4.9% per annum)
B-Class Tier 1 Capital (from PCA)	DKK 0.75 billion (base rate + 4.9% per annum)	DKK 0

Source: EC 2011.

Consolidation into FSC's Banking Group 1

In 2011, the FSC requested changes to the liquidation aid decision from the EC:

- i. an extended timeframe of the liquidation process [see Appendix B];
- ii. conversions of debt into equity in Roskilde to meet regulatory requirements;
- iii. the possibility to merge financial institutions in wind-down to achieve efficiencies and cost savings. (EC 2011)

The FSC agreed with the government that Roskilde's loss guarantee would be settled in 2011. The FSC would treat losses in connection with Roskilde in the same way that losses were handled under the Financial Stability Act (FSC 2011b). At the termination of Denmark's guarantee scheme on September 30, 2010, the FSC would appoint auditors to calculate profits and losses: equity at the end of the period minus capital contributions in subsidiaries plus market return (Danish Parliament 2008b, sec. 9.4, 17.4). If the FSC's financial statements showed a loss, the PCA guaranteed up to DKK 10 billion. If losses exceeded that amount, the PCA's annual guarantee commission of DKK 7.5 billion would be used. If the FSC's total losses exceeded these amounts, the PCA would further guarantee additional losses up to DKK 10 billion (Danish Parliament 2008b, sec. 9.5).

In 2011, the FSC restructured its holdings in assets of failed banks. As part of the restructuring, the FSC merged Roskilde Bank into an entity that it called FS Finans A/S (FSC 2012).

On July 1, 2012, the FSC remerged FS Finans A/S into the parent company (FSC 2013).

In 2012, also, the FSC allowed the state to pay off DKK 4.6 billion of the DKK 8.9 billion accumulated loss to FSC in return for a capital distribution from FSC (FSC 2013). In 2015 and 2018, the state paid off the final DKK 1.8 billion and DKK 2.5 billion in return for further capital distributions (FSC 2015; FSC 2017).

In February and October 2014, the FSC initiated sales processes to sell loans taken over from distressed banks, including loans from Roskilde (see Key Design Decision No. 11, Treatment of Assets) (FSC 2014a; FSC 2014b).

9. Treatment of Creditors and Equity Holders: Senior creditors, including depositors, were transferred to New Roskilde and paid in full. Subordinated creditors and equity holders remained with Old Roskilde and were to be compensated via an earn-out mechanism only after the satisfaction of all senior claims; the government considered it “highly unlikely” they would receive anything.

According to Roskilde’s half-year report in 2008, the bank recorded write-downs on loans and guarantees in the value of DKK 3.6 billion (Roskilde Bank 2008d).

The transfer of senior liabilities to New Roskilde constituted an event of default that enabled creditors to demand full redemption of their claims. Danish authorities anticipated that all senior creditors would make use of this right and used the credit facility to make these repayments by the end of October 2008. The EC noted that the main creditors of Old Roskilde were foreign banks (EC 2008b).

Old Roskilde’s investors in equity and subordinated or hybrid capital remained with the shell company that was emptied of all assets and other liabilities (EC 2008b; EC 2011). These investors were entitled to receive compensation through an earn-out mechanism—only after other claims were satisfied—which the FSC considered “highly unlikely” (see Figure 10) (EC 2011; FSC 2010; Roskilde Bank 2008d). Under the earn-out mechanism, any remaining funds after remuneration of the contributed share capital would go to the owners of subordinated capital and shareholders of Old Roskilde, in that order (Bernstein 2008; EC 2008b; Roskilde Bank 2008d). According to the EC, the use of an earn-out mechanism allowed Danish authorities to avoid two things. First, Danish authorities avoided a difficult valuation of Roskilde. Second, Danish authorities avoided allegations from investors that the business acquired by New Roskilde was mispriced (EC 2008b).

Figure 10: Roskilde Payout Order

Order	Creditors and Equity Holders
1	Senior liabilities (depositors, senior loans, and senior notes)
2	Tier 2 (capital provided by DNB)
3	Class A shares (capital provided by DNB)
4	Class B shares (capital provided by PCA, but shares later transferred to FSC for DKK 0)
5	Earn-out (owners of subordinated capital and equity)

Note: According to the EC's analysis, in the event of zero earn-out, investors in equity and subordinated capital would lose DKK 5.2 billion in book value, if comparing balance sheets from 2007 and August 29, 2008 (see Appendix A).

Sources: EC 2008b; EC 2011; FSC 2010; RNS 2008.

In December 2013, the FSC settled with the consumer ombudsman on the sale of bank shares by Old Roskilde. The settlement concerned private customers who bought shares in connection with two share sales campaigns: one from August 11, 2006, to September 17, 2006, and another from March 12, 2007, to April 8, 2007. The FSC identified 17,000 private customers entitled to damages. The FSC offered these customers compensation equivalent to 60% of their net losses (Danish Consumer Ombudsman 2013; FSC 2014c). According to the consumer ombudsman, the compensation took into account the general risk involved in buying Roskilde's shares and the additional risk of buying Roskilde's shares compared to other banks' shares. If shareholders did not want to accept this offer, the settlement said that the limitation period was suspended for 24 weeks after the settlement's conclusion. Thus, shareholders had the opportunity to submit a complaint to the FSC or Bank Appeals Board, or shareholders could file a court case. The settlement exempted Roskilde's senior employees, customer advisors, and others with special knowledge of the bank's conditions or special knowledge of the risk of buying shares (Danish Consumer Ombudsman 2013).

10. Treatment of Clients: Customers were transferred to New Roskilde and later to buyers of New Roskilde's assets. Complications arose with customer transfers that prolonged Roskilde's liquidation process.

The sale of 21 New Roskilde branches to Nordea, Spar Nord, and Arbejdernes Landsbank entailed the transfer of 90,000 customers, including depositors. Separation between deposits and loans was not allowed in the sales transaction. The FSC allowed the three buyers to return to the FSC any customer loans that they did not want to take over, at book value. Roskilde's buyers ultimately returned 6,000 retail customers and 1,400 corporate customers, thus complicating the timeline of New Roskilde's liquidation (see Key Design Decision No. 8, Approach to Resolution and Restructuring) (EC 2011; Roskilde Bank 2009). New Roskilde continued to serve customers on the usual banking terms but encouraged those customers to find new banks (Roskilde Bank 2009).

On December 10, 2008, to accelerate liquidation, the FSC announced that it was willing to fully or partially sell retail customers. The FSC sent details of the portfolio to 11 requesting

financial institutions. Of the 11 financial institutions, Arbejdernes Landsbank made an offer to assume the private customer portfolio for DKK 800 million, comprising 4,000 customers that were mostly retail (EC 2011; FSC 2011b). On March 31, 2011, the FSC and Arbejdernes Landsbank closed the deal; the private customer portfolio was not subject to debt recovery treatment and did not constitute a corporate commitment (EC 2011). Therefore, New Roskilde no longer had customers with deposit accounts only; New Roskilde still had 3,000 customers with outstanding commercial deposits because of their connection with loans (see Key Design Decision No. 11, Treatment of Assets) (Capital IQ; EC 2011).

In June 2011, when New Roskilde became a part of FS Finans, any remaining corporate customers were transferred to FS Bank. The FSC said that this reorganization would ensure uniform, continuous, and fair treatment of customers of all banks taken over by the FSC (FSC 2011b). FS Bank terminated its services and products in spring 2014. The FSC revoked FS Bank's license by the end of June 2014, at which point FS Bank became FS Finansselskabet. The FSA revoked FS Finansselskabet's license on September 30, 2014, suggesting no legacy corporate customers of Roskilde were serviced by the FSC after that time (FSC 2015).

11. Treatment of Assets: Assets were transferred to New Roskilde and later to buyers of New Roskilde's assets. More than 50% of corporate commitments were returned by buyers.

The sale of 21 New Roskilde branches to Nordea, Spar Nord, and Arbejdernes Landsbank entailed loan transfers. Separation between deposits and loans of individual customers was not allowed in the sales transaction. However, according to the transfer agreements, the three buyers could return loans acquired at book value during a six-week period;¹¹ contemporaneously estimated, 50% of the assets would be returned in the worst-case scenario (Dow Jones 2008b; EC 2011; Roskilde Bank 2009). In reality, more than 50% of corporate commitments were returned by the buyers (EC 2011).

Additional advances on existing loans were granted only in cases where they were necessary to preserve or increase the outlook of prepayment to New Roskilde or the relevant FSC company of outstanding loans. Some quantitative limits applied. For individual corporate clients, further advances could not exceed 20% on existing loans as of May 24, 2011. For corporate customers as a category at an aggregate level, further advances could not exceed 2.5% of New Roskilde's corporate loan portfolio (EC 2011).

In June 2011, the FSC reorganized legacy assets from Roskilde under the FSC subsidiary FS Finans (FSC 2011b; FSC 2012).

¹¹ According to a press release by Nordea, this bank could return "unattractive" loans and guarantees within six weeks of the completion of the transaction; Nordea could return loans to acquire shares in Roskilde within 24 months of the completion of the transaction (Nordea 2008). According to Spar Nord's annual report for 2008, the acquiring banks could return loans and guarantees whose credit qualities failed to meet the requirements of the acquiring banks; Spar Nord returned 300 business customers with a total business volume of DKK 700 million (Spar Nord 2009).

On February 28, 2014, the FSC launched a sales process to divest a customer loan portfolio consisting of exposures with 76 property companies with loans and credit facilities totaling DKK 1 billion. This loan portfolio included exposures from Roskilde among other distressed banks (FSC 2014a).

On October 1, 2014, the FSC launched a sales process to divest a customer loan portfolio consisting of 400 group customers (mostly Danish corporates) with loans and credit facilities totaling DKK 10 billion (FSC 2014b).

12. Treatment of Board and Management: Old Roskilde's board faced legal action for their trading activity and role in the bank's collapse. DNB and the PCA installed a new board at New Roskilde.

Management of New Roskilde

DNB and the PCA appointed new board members on August 29, 2008. DNB appointed chairman Henning Kruse Petersen (chairman of the FSC), Niels Heering (attorney at Gorrissen Federspiel Kierkegaard), and Visti Nielsen (board member of the FSC). The PCA appointed deputy chairman Jakob Brogaard (deputy chairman of the FSC) and Jørn Kristian Jensen (former managing director at Nordea Bank) (DNB 2008c; FSC 2010; FSC 2011b; Roskilde Bank 2009).

Søren Kaare-Andersen remained as managing director. Arne Wilhelmsen left as bank director, replaced by Lars Jensen (formerly a department director at Roskilde) (Roskilde Bank 2008c; 2009).

On September 12, 2008, a new board of directors was elected to New Roskilde at an extraordinary general meeting: chairman Jens Løgstrup, deputy chairman Jørgen Jensen, Erling Brønnum, Ole Nilensen, and Ole Scheel Krüger (Esmerk 2008b; Roskilde Bank 2008b).

Management of Old Roskilde

In early September 2008, former Roskilde City Councilor Rune Viberg took legal action against Old Roskilde's management (Dengsøe 2008; Esmerk 2008a). The charges included fraud, misleading accounting information, gross error and negligence, and providing incorrect information. Viberg specifically accused former bank director Wilhelmsen, former bank director Niels Valentin Hansen, and former deputy director Stig Bo Jensen of manipulating the share price because they sold their shares before Roskilde fell into trouble (Dengsøe 2008; Roskilde Bank 2007).

In March 2009, Danish media reported that former Roskilde chairman Peter Müller¹² signed a stock option agreement with former director Hansen after Roskilde's collapse and without the board's knowledge (Esmerk 2009).

¹² Müller was a board member of Roskilde since 1991; he was chairman of the board in 2008 before the bank's collapse (Roskilde Bank 2007; Roskilde Bank 2008d).

In March 2010, the Danish Securities Council and the DFSA cleared Roskilde's former board and management (DFSA 2009; Jyllands-Posten 2010). The council's report said:

The collective judgement is that both the bank's board and management at that time . . . expected that Roskilde Bank could survive the difficult situation as long as there was focus on liquidity and control of loans. No information exists that shows insider knowledge. (Jyllands-Posten 2010)

In 2010, the FSC brought a case against Roskilde executives for their role in the bank's collapse (Erhardsten 2022; Esmerk 2017; Ministry of Business and Growth 2013).

Subsequently, in November 2017, Denmark's Eastern High Court¹³ ruled that 10 Roskilde executives—including former director Hansen—would not have to pay DKK 1 billion in compensation costs (Esmerk 2017; FSC 2018). The court ordered the FSC to pay DKK 72.5 million in legal fees. The FSC appealed the decision with respect to the CEO and board of directors, and the auditors appealed the decision with respect the legal costs imposed on the FSC (FSC 2018; Supreme Court of Denmark 2022, Case No. 241/2017:4).

On December 1, 2022, the Supreme Court of Denmark ruled that former director Hansen was liable for actions resulting in Roskilde's collapse in 2008, thereby partially overruling the Eastern High Court (Supreme Court of Denmark 2022). The court ordered Hansen to pay a fine of DKK 231.7 million to the FSC within two weeks of the decision (Erhardsten 2022; Esmerk 2022; Sandøe 2022; Supreme Court of Denmark 2022, Case No. 241/2017:54, 60). Meanwhile, the court acquitted Roskilde's board members (Erhardsten 2022; Supreme Court of Denmark 2022, Case No. 241/2017:54).

13. Cross-border Cooperation: Danish authorities did not use the FSC until after the EC's approval. The EC approved some aid measures.

The DFSA had a cooperation agreement with Nordic authorities on the supervision of Nordea Group since 2000, which gave the DFSA powers to access Nordea's information and impose some regulatory requirements (IMF 2007). The sale of nine Roskilde branches to Nordea was subject to approval by the DFSA. The transaction closed on November 3, 2008 (Nordea 2009).

14. Other Conditions: The Danish government conditioned the unlimited credit facility on Roskilde Bank's sale to avoid moral hazard.

According to the NAO's report, the Danish government conditioned the unlimited credit facility on Roskilde's full or partial sale to avoid moral hazard—an expectation that other banks in a similar situation would expect a governmental safety net (DNB 2009; NAO 2009).

It is unclear whether salary restrictions on executives applied to Roskilde, since Bank Package I did not cover Roskilde, and Bank Package II (passed January 2009) later imposed

¹³ In Denmark, the hierarchy is as follows: (1) the Supreme Court, (2) the Eastern High Court and Western High Court, and (3) district courts (EU n.d.).

salary restrictions on executives who were subject to Bank Package I (FSC 2010; FSC 2011b; Ministry of Business and Growth 2013). Per Bank Package II, executives' variable salary components could not exceed 20% of the fixed base salary, including pension. Additionally, banks could not deduct more than 50% of executive board members' salaries in bank's tax accounts (Ministry of Business and Growth 2013).

15. Duration: Roskilde's liquidation took longer than expected.

The FSC's resolution of Roskilde took longer than Danish authorities anticipated because of complications with returned deposits and loans. In 2011, Danish authorities submitted a revised resolution timeline to the EC and requested changes to the liquidation aid decision. Danish authorities expected resolution to be complete by year-end 2012 (EC 2011).

The state wrote off the final debt related to its guarantee on losses incurred on the Roskilde Bank recapitalization in 2018 (FSC 2017).

By September 30, 2014, the FSA no longer had legacy customers from Roskilde (FSC 2015). It is unclear whether the FSC has remaining legacy assets from Roskilde.

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<https://ypfs.som.yale.edu/library/state-investor-last-resort-comparative-study-banking-crises-denmark-and-sweden>

Appendix A

Figure 11: Roskilde's Balance Sheet as of August 29, 2008, and Year-end 2007, DKK Millions

	29.08.2008	Year 2007
<i>ASSETS</i>		
Cash and funds in financial institutions	3 789	5 676
Loans	29 790	32 551
Bonds	4 359	2 599
Other tangible and intangible assets	1 307	1 157
Adjusted transfer value	-1 915	0
Total assets	37 330	41 983
<i>LIABILITIES</i>		
Debts to credit institutions and central bank	11 531	11 564
Deposits and other liabilities	17 691	17 376
Bonds issued	7 396	7 304
Other liabilities and provisions	712	581
<i>Total senior liabilities</i>	<i>37 330</i>	<i>36 825</i>
Subordinated capital	2 525	2 526
Equity	-2 525	2 633
Total equity and liabilities	37 330	41 983
<i>Off balance items</i>		
Contingent guarantees, etc.	9 258	11 895

Source: EC 2008b.

Appendix B

Figure 12: New Roskilde's Liquidation Process, as of May 2011

	<i>Factual</i>					<i>Projected</i>	
	YE 2007	31.10.2008, before sale	YE 2008, After sale	YE 2009	YE 2010 ⁹	YE 2011	YE 2012
Number of commercial customers¹⁰	14,425	13,617	5,422	4,076	2,714	1,610	1,265
Number of private customers	90,074	83,902	10,505	8,156	5,532	1,700 ¹¹	1,500 ¹¹
Commercial loans and guarantees, in nominal value, DKK million	32,154	30,508	27,924	21,138	17,128	15,200	9,700
Private loans and guarantees, in nominal value, DKK million	12,818	10,594	4,698	3,594	870 ¹²	n.a.	n.a.
Commercial deposits, DKK million	11,484	4,910	1,802	1,025	733	150	0
Private deposits, DKK million	6,406	4,127	363	294	78	n.a.	0
Total assets, DKK million	41,983	54,243	29,366	16,261	8,806		
Number of employees	654	612	234	209	203	150	80
Return before taxes, DKK million		n.a.	-3,501	-3,843	-2,327	-785	-202
Accumulated write-downs and provisions				10,272	10,381	9,700	5,600

Source: EC 2011.

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